Models — deep dive

The lifecycle

A model isn't a file you download once — it moves through a loop:

```
CHOOSE → REGISTER → SERVE → FINE-TUNE → EVALUATE → (loop back to serve)
                                (when needed)
   MLOps wraps it all: track · version · monitor
```

Two rules of thumb:

- Need new **knowledge**? → **RAG** (layer 3), not the model.
- Need new **behaviour**? → **fine-tune**. Train from scratch almost never.

Choosing — three dimensions

You pick the best model for a **task, a budget, and a GPU** — never "the best."

- **Open vs proprietary** — runs on your hardware vs vendor API; control & privacy vs frontier ceiling.
- **Size (LLM vs SLM)** — bigger reasons more generally; smaller is cheaper, faster, and often specialized.
- **Specialization** — reasoning · tool-calling · code · language · multimodal.

A fine-tuned 3B can beat a generic 70B on its one task — the case for SLMs.

The one number: VRAM

```
VRAM(weights) ≈ params × bytes   (fp16 = 2 · fp8 = 1 · int4 ≈ 0.5)
```

On our lab GPU — a free Kaggle **T4 (16 GB)**:

size	fp16	int4 (Q4)	fp16 fits?	Q4 fits?
3B	~6 GB	~2 GB	yes	yes
7–8B	~15 GB	~5 GB	tight	yes
14B	~28 GB	~8 GB	no	yes

On 16 GB, quantization is what lets a 7B+ fit with room for the KV cache.

Registry — what to track

Keep an honest catalog of the models in use. One table is enough; larger teams use MLflow Registry, the HF Hub, or a database.

Field	Why it matters
id / version	reproducibility — `model @ commit`
size & precision	drives the VRAM math
licence	commercial use? (next slide)
context length	how much it reads at once
capabilities	tools? vision? languages?
source & checksum	provenance + integrity

Licences — "open" is a spectrum

Open-weight ≠ open-source: you get the weights, rarely the training data/code.

Tier	Licence	You may…	Examples
Permissive	MIT · Apache-2.0	commercial use, fine-tune, redistribute, self-host — no strings	Qwen (Apache), DeepSeek (MIT), Mistral
Conditional	"community" · "modified MIT" · `license: other`	mostly — with limits (user caps, branding, approval)	Llama (community, 700M-MAU clause)
Restricted	custom / non-commercial	research & eval only	research-only releases

Economics: weights are free; vendors monetize API / enterprise / cloud — *commoditize the complement*. A model can start permissive and **tighten later** — read the licence every release.

Serving — engines

	Ollama / llama.cpp	vLLM
Best for	one user, demos	many users, throughput
Trick	GGUF + Q4 built in	PagedAttention + batching
Feel	snappy single answer	wins at concurrency

Latency = one fast answer · **throughput** = total tokens/s for everyone.

Two memory costs: fixed **weights** + a **KV cache** that grows with context × concurrency.

Serving — quantization

Fewer bits per weight: a little quality lost, a lot of memory saved.

Format	Bits	Where
fp16/bf16	16	training & high-fidelity serving
fp8	8	Blackwell/Hopper serving
GGUF Qk_*	~4–5	Ollama / llama.cpp
AWQ / GPTQ	4	vLLM / GPU serving

Lab 1: Qwen2.5-3B at fp16 vs Q4 on Ollama — Q4 ≈ ½ **VRAM**. On the T4 it's *not* faster (no int4 accel); on Blackwell it would be.

Fine-tuning — teach behaviour, cheaply

Full fine-tune of 7B = weights + grads + optimizer ≈ tens of GB → not on 16 GB.

- **LoRA** — freeze the base, train tiny adapters (<1% of params)
- **QLoRA** — load the base in **4-bit**, train the adapter on top

method	base	VRAM (7B)
LoRA	fp16/bf16	~18–24 GB
QLoRA	4-bit	~8–12 GB

You learn a few million adapter numbers, not the base weights — that's why a free 16 GB GPU can do it.

Lab 2: turn Qwen2.5-3B into a strict-JSON order parser; track in **MLflow**.

Evaluation — numbers, not vibes

A demo that "felt good" is not evidence. Measure before and after — and compare the base model against the tuned one.

kind	what
Benchmark	MMLU / GSM8K / ARC — general ability
Task eval	your held-out set + a metric you chose
LLM-judge / human	preference, helpfulness

Watch for **contamination** (memorized benchmarks), **quant quality** (measure it), and reporting a single number without speed and cost.

Lab 3: base vs fine-tuned on JSON-valid % / exact-match / field accuracy.

Training — rarely, and why

Pre-training from scratch = **trillions of tokens + a cluster** (that's why Layer 1 exists). Millions of dollars, months.

Justified only when:

- no existing model and no fine-tune can reach the goal
- a genuinely novel domain, modality, or language
- **continued pre-training** — keep training an open base on your corpus (the cheap middle ground)

Decision path, almost always: **use** → **fine-tune** → **(only then) train.**

MLOps — the glue

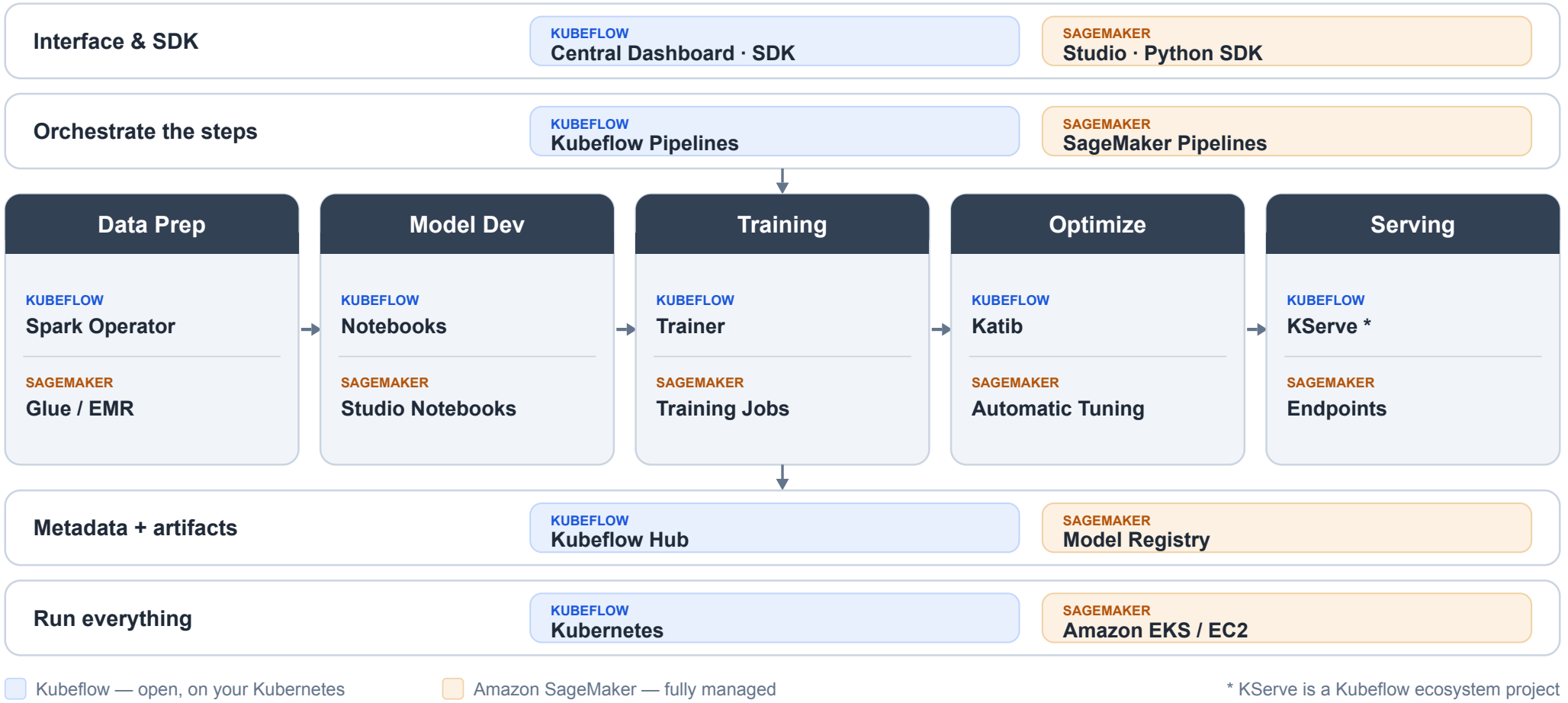
Wraps the whole loop: give every step a **versioned input and output** (reproducible), and **gate** the bad outputs before they ship.

Step	In → Out	MLOps
Register	need → model id + licence	catalog & version
Fine-tune	base + dataset → adapter + metrics	track run; version adapter + data
Evaluate	model + held-out set → scores	**gate: regression blocks deploy**
Serve	approved model → endpoint	monitor latency · quality · drift

The eval gate = "checkov blocks `terraform apply`", for models.

Systematic view — Kubeflow ≈ SageMaker

The seven phases are one **AI lifecycle** — two platforms implement it the same way:



If we built one — four forms of agency

Bostrom Ch. 10 — least → most autonomy. Maps onto how we ship LLMs today.

Form	Core risk	≈ today
Oracle — answers only	the answer itself manipulates	Q&A chatbot
Genie — one task, then stops	literal ≠ intended	task agent
Sovereign — open-ended, 24/7	no off-switch if misaligned	autonomous agent
Tool — no goals, you drive	optimization can still go agentic	function / tool call

Heuristic for **L4 orchestration**: grant the least autonomy that does the job, and keep a human in the approval loop.
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